

Highlights

Label-free covariance alignment for longitudinal EEG decoding

Esm E Moula Chowdhury Abha, Md Wahiduzzaman Suva, Muhammad Hasibur Rashid Chayon

- A 24,181-parameter EEG decoder with label-free alignment that adapts at test time.
- Accuracy is level with eight published decoders trained in the same pipeline.
- In-network alignment reduces session drift by 46–63% in 15 of 15 subjects.
- Test-time adaptation can track the streaming class and erase the signal it protects.
- A lower bound on adaptation rate, read from the protocol, predicted a new cohort.

Label-free covariance alignment for longitudinal EEG decoding

Esm E Moula Chowdhury Abha^a, Md Wahiduzzaman Suva^{a,*} and Muhammad Hasibur Rashid Chayon^a

^aDepartment of Computer Science, American International University-Bangladesh, Dhaka, Bangladesh

ARTICLE INFO

Keywords:

brain-computer interface
electroencephalography
test-time adaptation
covariate shift
selective prediction
calibration
lower-limb exoskeleton

ABSTRACT

A decoder that drives a powered lower-limb exoskeleton is fitted once and must keep working for weeks, while electrode impedance, montage and skin contact change the second-order statistics that every spatial filter depends on. Re-estimating the input covariance at inference and whitening by it, without labels, corrects this shift inside the network and removes the need for a calibration session. Such a layer carries a failure mode its offline counterpart cannot have. A running estimate tracks whatever is currently streaming, and under a protocol that presents classes in long contiguous blocks it converges to the active class and whitens away the signal it was inserted to protect.

We present a 24,181-parameter decoder built around such a layer, and derive a lower bound on its adaptation rate from the class-block timescale of the recording protocol. On a seven-participant exoskeleton cohort recorded across weeks it reaches 0.868 ± 0.026 balanced accuracy over three data-split seeds. Eight published decoders trained in the same pipeline average 0.817 to 0.878 over the same seeds, and no difference from this decoder is significant. The layer reduces session-to-session covariance distance by $45.9 \pm 4.9\%$ and $62.5 \pm 12.2\%$ on two cohorts, in 15 of 15 participants, against a control with no systematic reduction. On a treadmill cohort whose class blocks last 309 windows, the decoder transfers at 0.763 without the layer over three seeds, and enabling the layer costs 0.183. Slowing adaptation reduces that cost gradually, to 0.125 at a memory of 320 windows and 0.068 at 1600, and the layer does not become beneficial at any rate measured, while on the exoskeleton cohort slowing costs 0.050 over three seeds.

The lower bound then predicted a third cohort in advance. With class blocks of five windows, both predictions committed before any model was trained held: enabling alignment changed accuracy by +0.002, and slowing it by -0.006, where slowing reduces the cost of alignment on the treadmill cohort. On that cohort the decoder reaches 0.796 over three seeds, against 0.783 for the strongest published decoder, with expected calibration error 0.061 against 0.077 to 0.083. The upper bound, set by the drift timescale, could not be measured on these cohorts. Retraining after ICA artefact removal lowers accuracy by 0.051 and leaves it far above chance, and the losses do not follow the amount of muscle activity removed. A trained selective head adds no accuracy over three seeds and is reported as a negative result. The lower bound is a property of the experimental design, so it can be checked before a model is trained.

1. Introduction

Spinal cord injury leaves roughly 250,000 to 500,000 people per year with impaired or absent lower-limb function, and powered exoskeletons driven by non-invasive brain signals are one route to restoring overground mobility [1–3]. Decoding the wearer's intent to walk or to stop from scalp EEG is the control problem at the centre of such systems [4, 5]. It is not well described by per-epoch classification accuracy alone, for three reasons that shape the design presented here.

Longitudinal drift The decoder is fitted once and must keep working weeks later. In the primary cohort used here this is literal: the model is fitted on sessions 1–3 and evaluated on sessions 4–9, recorded across weeks. Electrode montage, impedance and skin contact all change between sessions, and the resulting shift in second-order statistics is the dominant source of non-stationarity in EEG decoding [6–8]. Session drift degrades fixed spatial filters and is the

reason most BCI systems require a per-session calibration block [9].

Asymmetric cost A false Walk command moves a person's legs. Balanced accuracy scores that identically to a missed Walk and cannot distinguish one sustained false activation from fifty scattered false windows. The quantity a wearer experiences is spurious activations per minute of standing, which we report alongside accuracy throughout.

Abstention A classifier trained only for accuracy must emit a decision for every window. A device that acts on a confidence value needs that value to be calibrated [10], and needs a principled way to do nothing when the evidence is weak [11, 12].

Existing responses to drift are mostly applied offline in preprocessing. Euclidean Alignment whitens each recording by the inverse square root of its mean covariance [13]; Riemannian Procrustes analysis and parallel transport generalise this to the manifold of symmetric positive-definite (SPD) matrices [14–16]. All of these estimate the alignment statistic over a complete recording, which means the target recording must exist before the decoder can run on it. Test-time adaptation removes that requirement in computer vision

*Corresponding author

✉ 26-94089-2@student.aiub.edu (E.E.M.C. Abha);

26-94088-2@student.aiub.edu (M.W. Suva); chayon@aiub.edu (M.H.R.

Chayon)

ORCID(s):

by updating normalisation statistics or parameters online [17–19], and recent work shows that such methods degrade when the test stream is temporally correlated instead of i.i.d. [20–22]. EEG data acquired under a blocked experimental protocol is temporally correlated by construction, which is the connection this paper makes precise.

Our contributions are:

1. We introduce an adaptive alignment layer that estimates and applies a whitening transform without labels and continues to update at inference, and we verify that its output reaches the classifier instead of being undone by the following normalisation (Section 4.1).
2. We identify a failure mode specific to in-network alignment: the running estimate tracks the currently streaming class when class blocks exceed the estimator’s memory, and the layer then removes the discriminative signal (Section 6.8).
3. We derive a lower bound on the adaptation rate from the class-block timescale of the protocol and test it prospectively on a third cohort, with both predictions and their decision thresholds committed before any model was trained on it (Condition 1, Section 6.10).
4. We validate the mechanism separately from the task, reporting a $45.9 \pm 4.9\%$ and $62.5 \pm 12.2\%$ reduction in session-to-session covariance distance in 15 of 15 subjects against a no-op control (Section 6.1).
5. We compare the decoder with eight published decoders trained in one pipeline on two cohorts, and report external validation on a treadmill cohort, where the decoder transfers and the alignment layer inverts the ordering of arms (Sections 6.4 and 6.5).
6. We retrain the decoder after ICA artefact removal. Most of the accuracy survives, and the losses that remain do not follow the amount of muscle activity removed (Section 6.3).

The decoder’s accuracy is level with the published decoders within the variation between seeds, and we make no claim of higher accuracy (Section 6.4).

2. Related work

2.1. Compact convolutional decoders for EEG

ShallowFBCSPNet and Deep4Net established that a square $\rightarrow$ pool $\rightarrow$ log pipeline over learned spatial filters recovers the band-power features that filter-bank common spatial patterns computes explicitly [23]. EEGNet reduced the parameter count by an order of magnitude using depthwise-separable convolutions [24], and EEGNeX refined the same family [25]. These architectures work well within a session and are the standard baselines in the field [26, 27]. Their limitation for our setting is that the learned spatial filters are fixed at training time, so a montage or impedance change between sessions is not compensated at all. We respond by placing an adaptive whitening stage before the spatial filters, so that what the filters see is

referred to a covariance estimated on the current recording (Section 4.1).

2.2. Attention-based decoders

EEG Conformer pairs a ShallowFBCSP-style stem with a transformer encoder [28], and EEG-TCNet applies temporal convolutions to the same problem [29]. The stem-plus-transformer combination is already published: an independent implementation of it in our pipeline reproduced EEG Conformer’s score without improving on it (0.848 against 0.843), and we therefore claim no novelty for that pairing. What these models share with the convolutional family is a fixed front end, which is the property the present work targets.

2.3. Alignment and transfer in BCI

Riemannian methods treat the spatial covariance as a point on the SPD manifold and classify with geodesic distances [30–32]. For transfer, Euclidean Alignment whitens each subject or session by $\mathbf{M}^{-1/2}$ where $\mathbf{M}$ is its mean covariance [13]; Riemannian Procrustes analysis adds rotation and scaling [14], and parallel transport moves covariances between tangent spaces [15, 16]. These methods are effective and label-free, and reviews report them as the strongest general-purpose transfer tools for EEG [33, 34]. Their limitation is operational. The alignment statistic is computed over a whole recording, so the recording must be complete before decoding starts, and within-recording drift is not tracked. We respond by moving the estimator inside the network as a layer that updates online, and we then show what that move costs (Section 6.8).

2.4. Test-time adaptation

Updating normalisation statistics on the test distribution improves robustness to corruption in vision [18, 19], and Tent extends this to updating affine parameters by entropy minimisation [17]. Test-time training adds a self-supervised objective at inference [35]. A recurring finding in this literature is that these methods assume the test stream is i.i.d., and degrade when it is not: NOTE addresses temporally correlated streams by maintaining a class-balanced reservoir [20], SAR stabilises adaptation under distribution shift with small or imbalanced batches [21], and CoTTA and LAME address error accumulation in continual adaptation [22, 36]. Our contribution relative to this line is quantitative: we show that in a blocked EEG protocol the relevant comparison is between the estimator’s effective memory and the duration of a single-class block, that this predicts the threshold location, and that the resulting constraint is two-sided because the same estimator is also what tracks the drift the layer exists to remove.

2.5. Calibration and selective prediction

Modern networks are overconfident, and expected calibration error [10, 37] degrades further under distribution shift [38], which is exactly the regime a longitudinally deployed decoder operates in. Selective classification allows a model to abstain [11, 12], and SelectiveNet trains the

classifier and the selection function jointly under a coverage constraint [39, 40]. We use SelectiveNet as published and claim no contribution to it. We include it because a decoder that must not move a wearer's legs on weak evidence needs a reject option, and Section 6.2 reports that the trained selection function did not provide a usable one.

2.6. EEG correlates of gait and exoskeleton control

Walking modulates cortical activity in a gait-phase-locked manner [41, 42], and treadmill walking can be decoded from scalp EEG [5, 43]. Robot-assisted gait modulates midline sensorimotor rhythms [44]. The intent signal our decoder targets is event-related desynchronisation in the mu and beta bands, a power decrease over sensorimotor cortex preceding and accompanying movement [45, 46]. This motivates a representation in which log-power ratios are preserved (Section 4.2), because ERD is defined as a relative power change.

3. Problem formulation

Let $\mathbf{X} \in \mathbb{R}^{C \times T}$ be a window of C channels and T samples with label $y \in \{\text{Stop}, \text{Walk}\}$. A recording session s supplies windows drawn from $p_s(\mathbf{X}, y)$. We assume the conditional $p(y | \mathbf{X})$ is stable across sessions but the marginal $p_s(\mathbf{X})$ is not, which is the covariate-shift assumption standard in this literature [6, 47]. Concretely we take the shift to act primarily on the spatial covariance $\Sigma_s = \mathbb{E}_{p_s}[\mathbf{X}\mathbf{X}^\top]/T$.

The decoder is fitted on sessions S_{fit} and evaluated on disjoint later sessions S_{test} , with no labels available from S_{test} at any point. This is the constraint that rules out per-session recalibration and motivates a label-free correction.

We quantify session drift by the affine-invariant Riemannian distance between SPD matrices,

$$\delta(\mathbf{A}, \mathbf{B}) = \left\| \log(\mathbf{A}^{-1/2} \mathbf{B} \mathbf{A}^{-1/2}) \right\|_F, \quad (1)$$

in preference to the Frobenius distance, because covariances do not form a vector space and the Frobenius distance is dominated by overall scale, which would flatter any whitening operation [30, 32].

Two timescales govern the analysis in Section 6.8. Let τ_{blk} be the median duration, in windows, of a contiguous single-class block in the recording protocol, and let τ_{drift} be the timescale over which Σ_s changes appreciably. Both are properties of the experimental design and the recording hardware, independent of the model.

4. Method

The decoder maps a window to a class posterior and a selection score. Its four components are shown in Fig. 1 with exact parameter counts.

4.1. Adaptive alignment

The layer maintains a running mean spatial covariance $\mathbf{M}$, applies $\mathbf{M}^{-1/2}$, and interpolates between the aligned and

the raw signal through a learned scalar gate $g \in [0, 1]$:

$$\mathbf{X}' = g \mathbf{M}^{-1/2} \mathbf{X} + (1 - g) \mathbf{X}. \quad (2)$$

Each window's covariance is normalised by its own trace before being accumulated, so that the estimate is an average of shapes instead of an average weighted by power:

$$\mathbf{M} \leftarrow (1 - m) \mathbf{M} + \frac{m}{N} \sum_{n=1}^N \frac{\mathbf{C}_n}{\text{tr}(\mathbf{C}_n)/C}, \quad \mathbf{C}_n = \frac{\tilde{\mathbf{X}}_n \tilde{\mathbf{X}}_n^\top}{T}, \quad (3)$$

where $\tilde{\mathbf{X}}_n$ is the mean-centred window, N the batch size and m the momentum. Without the trace normalisation the estimate is power-weighted. On a cohort whose majority class carries 1.52× the power of the minority class, the unnormalised estimate sits at Riemannian distance 0.306 from the majority-class covariance, that is, it is the majority class; the normalised estimate sits at 3.899 and 3.194 from the two class covariances, between them. Section 6.7 reports what correcting this did and did not achieve.

Two properties separate this layer from batch normalisation [48]. The update in (3) uses only the covariance of incoming windows and therefore runs on unlabelled test data. And the estimate continues to update in evaluation mode, whereas BatchNorm freezes its running statistics at test time. Freezing is correct when train and test share a distribution; here they are weeks apart, and freezing discards the adaptation the layer exists to provide.

Propagation of the aligned signal A whitening stage placed before a normalisation layer invites the objection that the normalisation simply undoes it. With identical initialisation on held-out windows, stem outputs computed with and without alignment differ by 36 % in relative magnitude and their correlation structure differs by 0.108, so the transform is not absorbed downstream.

4.2. Multi-scale power stem

Three parallel temporal resolutions (64, 128 and 256 ms), each with its own depthwise spatial filters, are followed by square $\rightarrow$ pool $\rightarrow$ log, after Schirrmeister et al. [23]. Amplitude normalisation precedes the squaring so that log-power ratios survive: scaling a window by s shifts its log-power by $2 \log s$, a constant that a later affine layer absorbs, whereas normalising after the square destroys the ratio in which ERD is defined [45].

This ordering is load-bearing, and the evidence is a dose-response across representations that differ in how much marginal power they carry (Table 6). Removing per-window amplitude normalisation is worth +0.134 and +0.174 on band-power features, which are pure marginal power; −0.016 and +0.029 on tangent-space features, which carry power only through the covariance diagonal; and −0.004 and +0.000 on a correlation-only representation, which is scale-free by construction and therefore cannot respond. A representation that carries no marginal power shows no effect, which is what identifies the mechanism instead of merely demonstrating it.

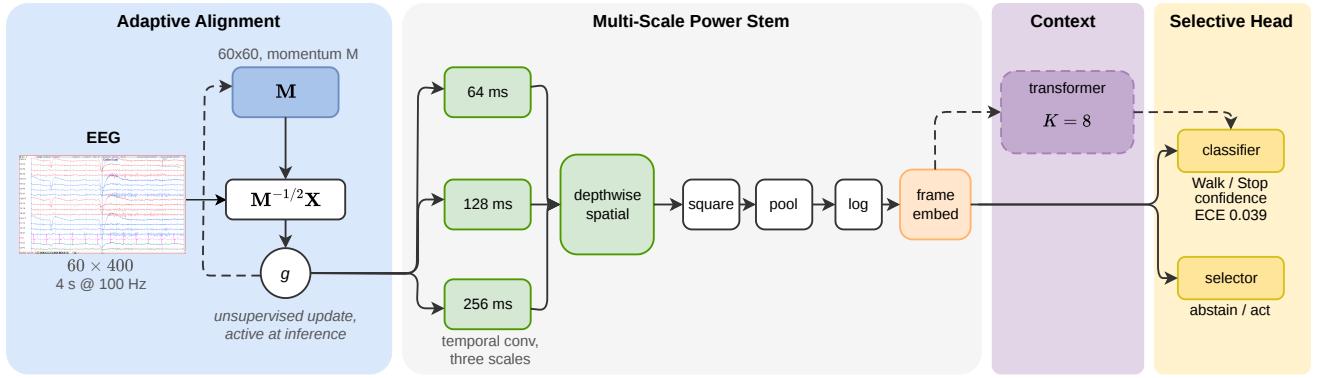


Figure 1: The decoder. The adaptive alignment layer maintains a running spatial covariance and whitens by it; its update path (dashed, left) uses no labels and remains active at inference, which is what separates it from a normalisation layer. The cross-epoch context pathway is optional and accounts for 594,816 of the 618,997 parameters of the full configuration, so the default configuration omits it and uses 24,181. Parameter counts are exact and sum to the total.

These are participant-paired comparisons, so they can be tested as such. Wilcoxon signed-rank tests over participants, Holm-corrected across the three representations within each cohort, give band power +0.174 (better in 7 of 8, $p_{\text{Holm}} = 0.047$) on cohort B and +0.134 (6 of 7, $p_{\text{Holm}} = 0.094$) on cohort A; tangent space +0.029 (5 of 8, $p_{\text{Holm}} = 0.500$) and -0.016 (2 of 7, $p_{\text{Holm}} = 0.750$); and the correlation-only control +0.000 (3 of 8, $p_{\text{Holm}} = 1.000$) and -0.004 (3 of 7, $p_{\text{Holm}} = 0.750$). **The effect reaches significance on the larger cohort and is null in the scale-free control, as the mechanism predicts;** cohort A is consistent in sign but underpowered at $n = 7$, where the smallest attainable Holm-corrected p is 0.047.

A separate sweep over where normalisation is applied in the network confirms the ordering at the architecture level: on cohort A with 4 s windows, per-window normalisation scores 0.849, global per-recording normalisation 0.878, and no normalisation 0.864, so the global setting used here is the best of the three and per-window normalisation is the worst.

4.3. Causal cross-epoch context

A causal transformer encoder [49] over a sequence of $K = 8$ epochs at stride 3 (a 14.5 s span), following the epoch-sequence formulation standard in sleep staging [50–52]. All masks are causal, so only present and past information is used, which is the only variant a worn device can run. Measured state persistence in cohort A supports a sequence model: the self-transition probability is 0.984 for Stop and 0.955 for Walk at a 0.5 s step, corresponding to dwell times of 12 to 32 s.

4.4. Selective head

A classifier paired with a selection function trained under a coverage constraint [39], with a full-coverage auxiliary term so that the backbone continues to learn from every sample, including the declined region. We use the formulation as published.

5. Experimental setup

5.1. Data

Cohort A (exoskeleton, OpenNeuro ds007788). Seven participants, nine sessions each, recorded across weeks while walking in a powered lower-limb exoskeleton [4]. Sixty-channel active EEG at 100 Hz; walk/stop labels are taken from the exoskeleton controller state. Windows of 4 s at 0.5 s stride give approximately 5031 fitting windows per participant. The model is fitted on sessions 1–3 and evaluated on sessions 4–9.

Cohort B (treadmill MoBI). Eight participants, three trials each, 64-channel EEG at 100 Hz, walk/stand from treadmill phase [43]. Fitted on trial 1 and evaluated on trials 2–3, with 2 s windows. The cohort is from an independent laboratory with different sensors, a different task framing, and reversed class balance: Stop is the minority class at approximately 12 % here and the majority class in cohort A.

Cohort C (motor execution, PhysioNet EEGMIDB). Twenty participants, S001–S020, from the PhysioNet motor movement and imagery database [53, 54], recorded with 64-channel EEG at 160 Hz and resampled to 100 Hz. Only the motor-execution runs are used, and the task is rest against movement of the hands or feet. The model is fitted on runs 3, 5, 7 and 9 and evaluated on runs 11 and 13, with 2 s windows at 0.5 s stride, each lying wholly inside one annotated period. The protocol alternates the two classes about every 4 s, a median of 5 windows per class block. Its analysis plan was committed to version control before any model was trained on it (Section 6.10).

The protocols differ sharply in temporal structure, which Section 6.8 shows to be decisive (Table 1).

5.2. Preprocessing and training

Signals are band-pass filtered to 8–30 Hz and resampled to 100 Hz. Cohort A uses a zero-phase FIR filter in MNE-Python [55], after which the EOG channels, filtered to the same band, are regressed linearly out of the EEG. Cohorts B and C use a zero-phase fourth-order Butterworth filter and have no EOG step. No other artefact rejection, ICA

Table 1

Contiguous single-class block structure. τ_{blk} is the median block duration in windows.

Cohort	Blocks	τ_{blk}	Longest
A (exoskeleton)	109	18	126
B (treadmill)	7	309	2211

Table 2

Training configuration, identical across all arms, all cohorts and the published decoders of Table 3, except where the cohort definition differs.

Setting	Value
Band-pass / resample	8–30 Hz, 100 Hz (zero-phase)
Artefact correction	EOG regression (A); none (B, C)
Amplitude normalisation	global, per recording
Window / stride	4 s / 0.5 s (A); 2 s (B); 2 s / 0.5 s (C)
Context length K , stride	8 epochs, 3 (14.5 s span)
Optimiser	AdamW [56], lr 3×10^{-4} , weight decay 10^{-2} , one-cycle schedule
Batch size N	32
Epochs	at most 30; early stopping after 12 without improvement
Gradient clipping	norm 1.0
Loss	cross-entropy, inverse-frequency weighted
Selective coverage target	0.90
Adaptation momentum m	0.2 (swept in §6.8)
Model selection	best held-out balanced accuracy
Calibration	temperature scaling fitted on the held-out split

decomposition or channel interpolation is applied, so that the alignment layer is evaluated on the shift it is meant to correct. The band also excludes the gait cycle, near 1–2 Hz, and most electromyographic power above 30 Hz. Amplitude normalisation is applied globally per recording, before the squaring stage, for the reason given in Section 4.2.

Table 2 lists the training configuration. The adaptation momentum is $m = 0.2$ unless stated otherwise; Section 6.8 treats m as the object of study. Model selection uses balanced accuracy on a held-out split of the fitting sessions, evaluated at the best epoch by that criterion, which is conventional but upward-biased on small folds (Section 8).

Published decoders Eight published decoders from the *braindecode* library [57] were trained in the same pipeline as the proposed decoder: ShallowFBCSPNet and Deep4Net [23], EEGNet [24], EEGNetX [25], EEG Conformer [28], TSception [58], FBLightConvNet and EEG-TCNet [29]. Each model's classification layer is replaced by the same linear projection and classifier, and each is trained with the preprocessing, optimiser, schedule, early stopping, model selection and temperature scaling of Table 2, so that the models differ only in the backbone. No per-model tuning was

performed. Parameter counts include the shared classifier and are instantiated at cohort A's input shape.

5.3. Outcome measures

Balanced accuracy for comparability with the literature; expected calibration error (ECE) with 10 equal-width bins [37], because a device acting on a confidence value needs that value to mean something; accuracy at 90 % coverage, the trained selective operating point; spurious activations per minute of standing; and detection latency, reported alongside the benefit it buys.

Leakage guard Unconditioned nuisance-decodability is confounded with accuracy. In our pipeline a decoder that uses no movement information whatever measures $R^2 = +0.863$ against the movement signal, because both the decoder output and the nuisance are driven by the class label. Movement leakage is therefore measured conditional on the class label throughout, by centring the nuisance features within class before regression.

We calibrate the probe by injecting the recorded movement signal into the EEG, spatially mixed with a fixed per-participant gain matrix, at a known fraction f of signal power. The calibration uses the tangent-space representation. Accuracy rises monotonically with f : 0.763 at $f = 0$, then 0.819, 0.873, 0.892 and 0.913 at $f = 0.001, 0.005, 0.01$ and 0.05 . The conditional probe rises with it, from -0.112 at $f = 0$ through $+0.015$ at $f = 0.005$ to $+0.067$ at $f = 0.02$, where accuracy is 0.903. Reaching an accuracy near 0.90 through injected movement therefore moves the probe to about $+0.07$. The proposed *align + gate* decoder reaches 0.901 with a conditional probe of -0.068 , and all eight cohort-A arms lie between -0.066 and -0.082 , on the uncontaminated side of the calibration curve. Two limits apply. The calibration was performed on a different representation from the decoder it is compared with, and the conditional probe cannot register artefact information that is constant within each class. The comparison is therefore evidence against movement-driven accuracy. Section 6.3 adds a second, independent test by removing artefact components and retraining.

5.4. Statistical treatment and measurement noise

Drift reductions are tested with a two-sided paired t -test over participants, pairing each participant's aligned distance against its raw distance. Accuracy comparisons are reported as the mean over participants with the between-subject standard deviation, and as paired Wilcoxon signed-rank tests over participants wherever per-participant results were retained (Section 8).

Two ablation arms disable alignment entirely and therefore never invoke the running estimator. For them the momentum and the estimator version are inert, so every run is a true repeat and they measure the noise floor directly (Table 10).

The stem-only arm reproduces 0.7375 bit-for-bit across three runs; the arms containing the cross-epoch transformer

Table 3

The proposed decoder and eight published decoders trained in one pipeline, with identical preprocessing, optimiser, schedule, early stopping, model selection and classifier head. Balanced accuracy (Acc) and expected calibration error (ECE). Cohort A has seven participants and cohort C twenty; cohort C values are means over three data-split seeds. Parameter counts include the shared classifier and are instantiated at cohort A's input shape. Seed-0 differences on cohort A lie within the proposed decoder's range across seeds and are not ranked. † near chance (< 0.55) for at least five participants on a seed, a training failure under the shared settings. **Red cells await runs in progress.**

Model	Params	A Acc (seed 0)	A ECE (seed 0)	A Acc (3 seeds)	C Acc	C ECE
<i>This work</i>						
Ours, align + gate	24 181	0.884	0.042	0.868 ± 0.026	0.796	0.061
<i>Published decoders, same pipeline</i>						
EEGTCNet [29]	7 062	0.871	0.077	0.878 ± 0.009	0.585†	0.054
FBLightConvNet	23 760	0.870	0.059	0.866 ± 0.011	0.778	0.077
ShallowFBCSPNet [23]	206 962	0.870	0.068	0.849 ± 0.020	0.765	0.078
EEGNet [24]	27 666	0.868	0.062	0.856 ± 0.028	0.630†	0.059
EEGNetX [25]	71 010	0.862	0.068	0.876 ± 0.014	0.783	0.083
EEGConformer [28]	445 442	0.857	0.066	0.847 ± 0.008	0.783	0.077
TSception [58]	182 098	0.856	0.076	0.852 ± 0.003	0.723	0.079
Deep4Net [23]	717 417	0.823	0.058	0.817 ± 0.014	0.549†	0.139

do not, and the attention path is the only structural difference between them. We attribute the variation to non-deterministic reductions in scaled dot-product attention on GPU.

Run-to-run variation at a fixed split seed is a lower bound on the relevant noise. Across three data-split seeds the proposed configuration spans 0.839 to 0.884, a range of 0.046 and a standard deviation of 0.026 (Table 10), several times the fixed-seed figure. This is the quantity that governs whether two models can be ranked, and differences between models smaller than it are reported without a ranking. The spread also differs between arms. On cohort B at a memory of 320 windows it is 0.003 for *align + gate* and 0.037 for the no-align arm, which contains the transformer, so single-seed values of the no-align arm on that cohort lie inside its own seed spread.

Any arm containing the transformer carries approximately 0.010 run-to-run noise at a fixed seed, so differences below 0.02 are not interpreted anywhere in this paper. The effects we do interpret on accuracy are measured over three data-split seeds and tested paired within participant (Tables 3 and 5, Section 6.9). An exact null in the deterministic arm is stronger evidence than one merely within noise, because that arm has no noise to hide in.

Pipeline separation The proposed decoder and the published decoders of Section 6.4 were trained in one pipeline. Tables 6, 7 and 8 come from earlier pipelines and are used only for comparisons within each table.

6. Results

6.1. The drift mechanism

We measure the Riemannian distance (1) between the fitting-session mean covariance and each held-out session's, before and after the layer (Fig. 2, Table 9).

Drift is reduced in 15 of 15 participants ($t = -9.46$, $p = 8 \times 10^{-5}$ for cohort A; $t = -12.33$, $p = 5 \times 10^{-6}$ for cohort B), with per-participant reductions spanning 38.7 to 51.2 % on cohort A and 43.8 to 82.1 % on cohort B (Table 9). No participant regresses, and the reduction is not carried by one or two outliers. The no-op control leaves the distance unchanged, which establishes that the effect comes from adaptation and excludes both the whitening operation and the measurement itself.

The no-op control runs at momentum 0. Its running estimate is primed by copying the first batch's covariance and never updates, so it applies one fixed linear map to every session, $M = gW + (1 - g)I$ with $g = \sigma(2.0) \approx 0.88$. A fixed map applied to both covariances is a congruence, under which the affine-invariant distance (1) does not change, and the control accordingly changes the distance by less than 0.1 % of its raw value in every participant (Table 9). An earlier version of this measurement normalised each covariance by its trace after the map, which breaks that invariance and made the control scatter in both directions; it was removed, and the values reported here replace the larger reductions it produced (52.0 and 83.9 %).

The mechanism works as designed, and Section 6.5 shows that this does not settle whether it helps the task.

6.2. Component ablation, cohort A

The acc@90 value for the *align + ctx* arm is withheld because with the selection head disabled the model emits near-constant confidence, so ranking by it selects the first 90 % of the test set in recording order. That number would be an ordering artefact.

Selective prediction Abstention is supposed to raise accuracy on the windows that are kept. It does not. Across the 18 arms in this work for which both quantities were computed on the same windows with the same metric, declining the

Table 4

The adaptation-rate condition, single seed. Momentum m sets the estimator's memory to roughly $32/m$ windows: 160, 640 and 3200. **Every arm that uses alignment gains as the memory crosses the class-block length, and neither arm that omits it responds;** the stem-only arm reproduces 0.737 exactly at all three settings despite a 160-fold change. The lower block applies the identical change to the cohort whose blocks are short, where it costs 0.060, so the two cohorts have opposite optima. Three-seed values for cohort B are in Table 5, and for cohort A in Section 6.9.

Arm	$m=0.20$	$m=0.05$	$m=0.01$	Δ at crossing	Aligns	Δ cohort A
<i>Cohort B ($\tau_{\text{blk}} = 309$): slowing adaptation past the class block</i>						
align + ctx + gate	0.631	0.724	0.723	+0.093	yes	—
align + gate	0.581	0.671	0.696	+0.090	yes	—
align + ctx	0.606	0.700	0.726	+0.094	yes	—
ctx + gate	0.788	0.782	0.783	-0.007	no	—
stem only	0.737	0.737	0.737	+0.000	no	—
<i>Cohort A ($\tau_{\text{blk}} = 18$): the same change, matched estimator</i>						
align + ctx + gate	0.862	—	0.801	-0.060	yes	-0.060
ctx + gate	0.834	—	0.827	-0.007	no	-0.007

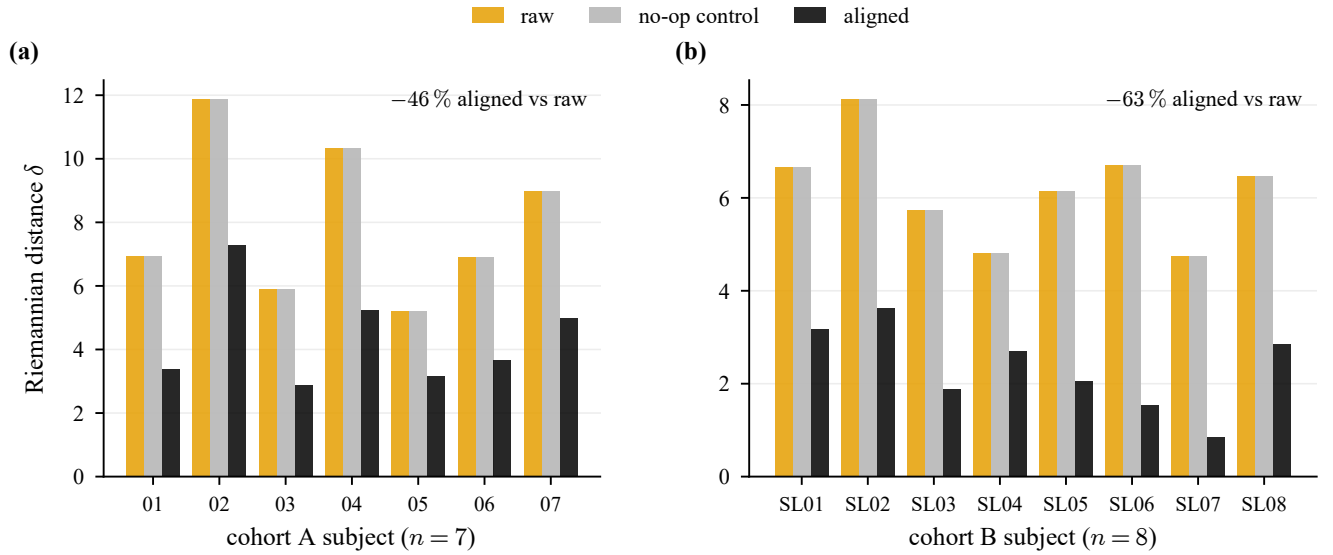


Figure 2: Session-drift reduction per participant. Bars are the Riemannian distance δ between the fitting-session mean covariance and the held-out session mean, for the raw signal, the momentum-0 no-op control, and the aligned signal. (a) cohort A, $n = 7$; (b) cohort B, $n = 8$. Distance falls in every participant of both cohorts; the no-op control, which shares every code path but never updates its estimate, leaves the distance unchanged.

least-confident 10 % **lowers** balanced accuracy in 14, by 0.045 on average; the headline arm loses 0.026 (0.901 $\rightarrow$ 0.876) and the worst case loses 0.122. The four arms where abstention helps are exactly those with no trained gate (*align* + *ctx*, *stem only*), where the ranking is arbitrary. A gate that is beaten by its own absence is not ranking windows by correctness, and the selective head is therefore reported as a component for which no benefit was demonstrated. Fixing it, for instance by thresholding a temperature-scaled softmax confidence instead of the learned gate, is left open. Calibration is compared with the published decoders in Section 6.4.

The head contributes no accuracy either. Under the trace-normalised estimator, an *align*-only arm reaches 0.883 against 0.884 with the head, a difference of 0.002, and adding the head to *align* + *ctx* changes accuracy by 0.000 (0.862

in both arms). Calibration does not suffer either: without the head ECE is 0.038, against 0.042 with it, both under the same estimator. Over three data-split seeds the *align*-only arm averages 0.869 against 0.868 with the head, a difference of -0.001 (higher with the head in 4 of 7 participants, Wilcoxon $p = 1.00$), and mean ECE is 0.054 without the head against 0.058 with it. An earlier difference of +0.016 between the same two context arms under the original estimator is 1.6 $\times$ the 0.010 noise floor of arms containing the transformer (Table 10), and it does not reappear in the comparison made with one estimator throughout. On this evidence the selective head could be removed with no loss of accuracy or calibration. It is kept in the arms reported here so that they remain comparable with the ablation.

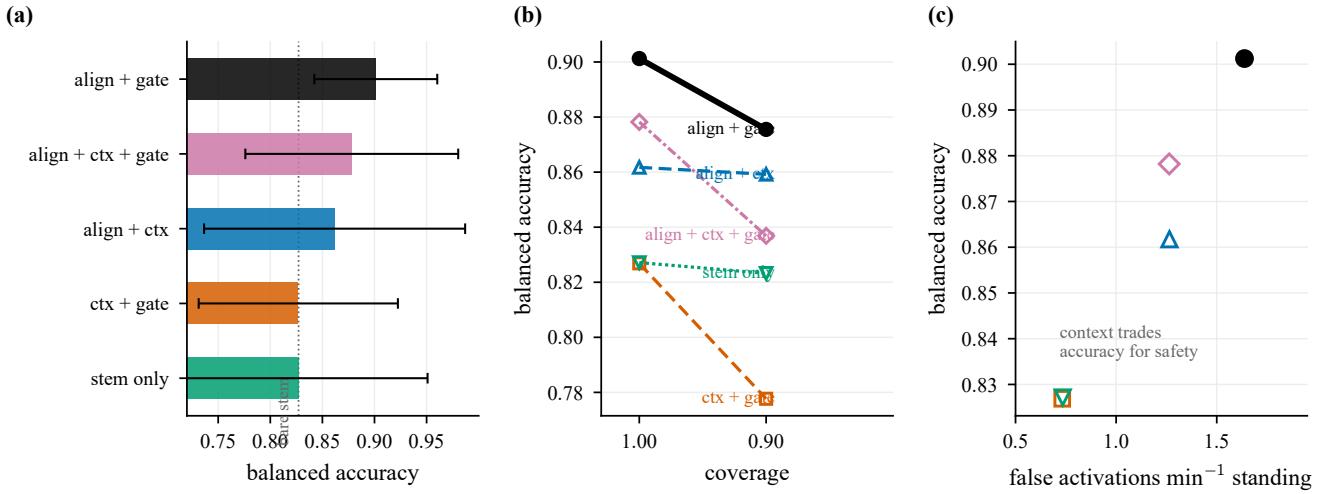


Figure 3: Component ablation on cohort A, seed 0. (a) Balanced accuracy by arm; error bars are ± 1 SD across the 7 participants; seed variation is reported separately in Table 10. (b) Balanced accuracy at 90 % coverage, obtained by declining the least-confident 10 % of windows. It sits *below* full-coverage accuracy for every arm with a trained gate. (c) Balanced accuracy against spurious activations per minute of standing, showing the cross-epoch context as a selectable operating point. [To be regenerated with three seeds.]

Table 5

Cohort B over three data-split seeds under the current estimator. *Cost* is *align + gate* minus the no-align arm, which does not use the estimator and is averaged over its 6 runs (0.763), paired within participant with seeds averaged; *p* is the Wilcoxon signed-rank test. The cost falls gradually as the memory lengthens, with no step at $\tau_{\text{blk}} = 309$, and it stays negative at every memory measured.

<i>m</i>	Memory	align + gate	Cost	Lower in	
0.20	160	0.580 $\pm$ 0.008	-0.183	7 of 8	0.016
0.10	320	0.638 $\pm$ 0.002	-0.125	7 of 8	0.026
0.02	1600	0.695 $\pm$ 0.005	-0.068	6 of 8	0.148
0.01	3200	pending	pending	pending	pending

Dependence on the alignment stage With alignment removed, the context and selection modules together contribute between 0.000 and 0.007: the no-align arm spans 0.827–0.834 across repeats (Table 10) while the stem-only arm is deterministic at 0.827. Against the 0.010 noise floor that is indistinguishable from nothing, and well below alignment’s own contribution: +0.051 under the original estimator (0.878 against 0.827, Table 11) and +0.028 under the trace-normalised one (0.862 against 0.834) from the estimator-fix run of Section 6.7. The lower figure should be read with care: the no-align arm is inert to the estimator, so its 0.834 differs from 0.827 only by run-to-run variation, and part of the gap between the two figures is that noise. The comparison that isolates alignment most directly, *align + gate* against *gate* with nothing else changed, was repeated over three data-split seeds under the trace-normalised estimator. With each participant’s accuracy averaged over seeds, alignment raises accuracy by +0.026 (0.868 against 0.842), higher in 6 of 7 participants (Wilcoxon $p = 0.22$). The effect

Table 6

Two-condition comparisons. **The first block is the identifying evidence for the stem design:** removing per-window normalisation is worth +0.134 and +0.174 where the representation is pure marginal power, and **exactly +0.000 where it is scale-free by construction** and therefore cannot respond. An effect that scales with power content and vanishes without it identifies the mechanism instead of merely demonstrating it. The second block retired an amplitude side-channel: supplying a per-window amplitude feature and supplying the same feature with its window assignment shuffled differ by under 0.02 for four of five models.

Comparison	Power content	Condition	Control
<i>Per-window amplitude normalisation, by how much marginal power the re</i>			
band power, cohort A	full	0.625	0.760
band power, cohort B	full	0.574	0.748
tangent space, cohort A	partial	0.762	0.746
tangent space, cohort B	partial	0.675	0.704
correlation, cohort A	none	0.738	0.734
correlation, cohort B	none	0.701	0.702
<i>Amplitude side-channel against its shuffled control</i>			
ShallowFBCSPNet	—	0.832	0.816
Deep4Net	—	0.798	0.809
EEGNeX	—	0.841	0.841
EEGITNet	—	0.828	0.837
BDTCN	—	0.748	0.652

is +0.067, +0.029 and -0.019 on the three seeds, and it is uneven across participants: sub-04 gains 0.179 and sub-05 loses 0.127, each with the same sign on every seed. The gain from alignment on cohort A is therefore close to the trace-normalised single-seed figure of +0.028, and with seven participants it is not significant. The modules do not stack; they depend on the alignment stage to be useful at all.

Table 7

Where amplitude normalisation is applied, across 6 published decoders on cohort A, single seed, in an earlier training pipeline. All three columns are balanced accuracies; best per row in bold. Per-window normalisation is best for 2 and worst for 2, and most differences are below the 0.02 noise threshold, so this comparison neither supports nor contradicts the representation-level result in Table 6.

Model	Per-window	Global	None
ShallowFBCSPNet [23]	0.824	0.821	0.818
Deep4Net [23]	0.760	0.821	0.821
EEGNeX [25]	0.834	0.832	0.835
EEGITNet [59]	0.833	0.791	0.795
BDTCN	0.820	0.822	0.820
EEGConformer [28]	0.828	0.829	0.805

Cross-epoch context Adding it costs 0.023 accuracy and raises ECE from 0.039 to 0.074, while reducing spurious activations from 1.64 to 1.26 per minute, a 23 % reduction. It is therefore reported as a selectable safety mode, outside the default configuration. Removing it also removes 594,816 of the model's 618,997 parameters, which is what makes the default configuration compact.

6.3. Artefact control

Walk and stop differ in movement, so accuracy on this task could arise from muscle or motion artefact. The proposed configuration was therefore retrained after independent component analysis, with both arms restricted to the training task so that they differ only in the cleaning step. Each recording was re-referenced to the average, a copy band-passed to 1–45 Hz was decomposed into 30 components with the extended Picard algorithm [60], and the components that ICLabel [61] classified as eye, muscle, heart, line noise or channel noise were removed before band-passing to the decoding band. Across the 63 recordings, a mean of 6.8 of 30 components was removed, most often eye (42 % of removals) and muscle (38 %). Gait-related artefact is difficult to separate from cortical activity during walking [62, 63], and ICLabel is not trained on it specifically, so this control narrows the question and leaves part of it open.

Cleaning lowers balanced accuracy from 0.835 to 0.785, a paired difference of -0.051 (higher in 2 participants, lower in 5; Wilcoxon $p = 0.22$; Table 12). Most of the accuracy therefore survives, well above the 0.500 chance level, and four participants are essentially unchanged or improve. The loss is concentrated in three participants, of 0.11 to 0.16. It does not follow the amount of muscle activity removed. The participant with the most muscle components removed, 7.1 per recording, improves by 0.016, and one with 0.3 removed loses 0.124 (Spearman $\rho = +0.22$ between muscle components removed and accuracy change). Dependence on muscle artefact would produce the opposite pattern. Two of the three largest losses fall on the participants with the lowest baseline accuracy, 0.673 and 0.680, where estimates are noisiest. The conditional leakage probe stays negative

in both arms (-0.088 and -0.077). With seven participants, and with re-referencing also part of the cleaning, the control falls short of establishing that the accuracy is neural, and it gives no support to the view that the accuracy rests on muscle artefact.

6.4. Comparison with published decoders

Table 3 compares the proposed configuration with the eight published decoders of the experimental setup, all trained in the same pipeline.

On cohort A, over the same three data-split seeds, the proposed configuration averages 0.868 and the published decoders average 0.817 to 0.878. Paired over participants with seeds averaged, no difference is significant in either direction. EEG-TCNet and EEGNeX lie 0.010 and 0.008 above it (higher in 4 and 3 of 7 participants; Wilcoxon $p = 0.58$ and $p = 0.94$), and the other six lie 0.003 to 0.051 below it (smallest $p = 0.22$). At the first seed alone the proposed configuration is the highest of the nine (0.884 against 0.823 to 0.871), so single-seed orderings on this cohort are not stable. It holds no parameter advantage over the smallest published decoders: EEG-TCNet averages 0.878 with 7,062 parameters and FBLightConvNet 0.866 with 23,760, against 24,181.

On cohort C, over three seeds, the proposed configuration averages 0.796, and the published decoders that trained reach 0.723 to 0.783. Paired over participants with seeds averaged, it is higher than EEGNeX by 0.013 (higher in 15 of 20 participants; Wilcoxon $p = 0.17$), than EEG Conformer by 0.013 (12 of 20; $p = 0.15$), than FBLightConvNet by 0.018 (15 of 20; $p = 0.10$), than ShallowFBCSPNet by 0.031 (16 of 20; $p = 0.011$) and than TSception by 0.073 (19 of 20; $p < 0.001$). After Holm correction across these five comparisons, only the differences from ShallowFBCSPNet ($p = 0.044$) and TSception ($p < 0.001$) remain below 0.05. Its expected calibration error is 0.061, against 0.077 to 0.083 for the same decoders. EEGNet, Deep4Net and EEG-TCNet did not train reliably on this cohort under the shared settings: on each seed they stay near chance (< 0.55) for 6 to 9, 10 to 12 and 7 to 11 of the twenty participants respectively, so their cohort C values record a training failure. The low calibration error of EEGNet and EEG-TCNet there (0.059 and 0.054) shows why calibration error is read together with accuracy, since a decoder near chance that reports low confidence is well calibrated.

On cohort A the proposed configuration's calibration error averages 0.058 across seeds, against 0.058 to 0.076 for the published decoders over the same seeds. Its accuracy is therefore level with the published decoders on cohort A and slightly higher on cohort C, without a significant difference against the strongest of them, and its calibration error is lower on cohort C. What the architecture adds at this accuracy is an alignment layer that updates without labels at test time, a measured reduction of session drift (Section 6.1), and a rule, checkable before training, for when that layer must be slowed (Section 6.8).

Table 8

Every architectural intervention tested, with the control that decided it. The motion-canceller row is the clearest rejection: the module scored 0.905 with the inertial channel present and chance with it zeroed, so the module was classifying the inertial signal. The temporal-smoothing rows are the clearest retention: shuffling window order removes the gain and drops accuracy **0.113 below the unsmoothed baseline**, which no operation helping for an unrelated reason would do. n is seeds.

Intervention	n	With	Control	Δ	Verdict
<i>Rejected: the control removed the effect</i>					
Motion cancellation	1	0.905	0.501 (zeroed)	-0.404	inertial classifier
Motion cancellation	1	0.905	0.529 (shuffled)	-0.376	same
Within-epoch attention	3	0.840	0.863 (removed)	+0.024	block not earning its place
MRCP dual-band pathway	1	0.742 (both)	0.775 (ERD only)	+0.034	low band adds noise
Capacity scaling	1	0.768	0.608	+0.159	24 configs, no trend
Alignment (SPD backbone)	3	0.823 (EA)	0.821 (none)	-0.002	inside seed spread
<i>Kept: the control preserved the effect</i>					
HMM forward filter	4	0.802	0.778 (baseline)	+0.024	survives shuffle
Viterbi decoding	4	0.817	0.778 (baseline)	+0.038	survives shuffle
control: shuffled order	4	0.665	0.778 (baseline)	-0.113	falls below baseline

Table 9

Per-participant session drift. δ is the affine-invariant Riemannian distance between the fitting-session mean covariance and the held-out mean. The no-op control shares every code path but never updates its estimate. **Every participant improves:** 15 of 15, $t = -9.46$, $p = 8 \times 10^{-5}$ (A) and $t = -12.33$, $p = 5 \times 10^{-6}$ (B). The control changes the distance by at most 0.09% of its raw value.

Participant	Raw δ	No-op	Aligned	Reduction
<i>Cohort A, 6 held-out sessions</i>				
A01	6.925	6.926	3.385	51.1%
A02	11.867	11.878	7.273	38.7%
A03	5.891	5.891	2.875	51.2%
A04	10.337	10.339	5.240	49.3%
A05	5.208	5.208	3.163	39.3%
A06	6.909	6.909	3.650	47.2%
A07	8.986	8.987	4.986	44.5%
<i>Cohort B, 2 held-out trials</i>				
SL01	6.658	6.658	3.176	52.3%
SL02	8.122	8.126	3.623	55.4%
SL03	5.739	5.739	1.879	67.3%
SL04	4.803	4.803	2.701	43.8%
SL05	6.153	6.154	2.050	66.7%
SL06	6.696	6.696	1.546	76.9%
SL07	4.737	4.737	0.846	82.1%
SL08	6.462	6.462	2.854	55.8%

6.5. External validation and the inverted ordering

The decoder transfers: without the alignment layer it reaches 0.763 on cohort B averaged over three seeds (0.792 at seed 0 in Table 13), against a majority-class floor of 0.500 balanced accuracy on a cohort whose classes split 87.7/12.3. What does not transfer is the alignment layer. The best cohort-A configuration is the worst cohort-B one, and the ordering inverts, with all three arms that use alignment falling below both arms that do not. The layer is worth +0.074 on cohort A (0.901 against 0.827) and -0.183 on

Table 10

Measurement noise. The first block repeats arms for which momentum and estimator version are inert, so all variation is run-to-run: the arm without the cross-epoch transformer is bit-identical across three runs, and the arms with it are not. Across data-split seeds the spread is larger again, **0.032 to 0.046** for our arms without the transformer. **Differences below 0.02 are not interpreted anywhere in this work**, and differences below the seed spread are reported without a ranking.

Arm	Transformer	n	Spread
<i>Fixed seed, repeated runs of arms where the manipulated variable is inert</i>			
stem only, cohort B	no	3	0.0000
no-align, cohort A	yes	3	0.0069
no-align, cohort B	yes	4	0.0107
<i>Across three data-split seeds, trace-normalised estimator</i>			
align + gate	no	3	0.0455
gate only	no	3	0.0409
align only	no	3	0.0320
<i>Cohort B, memory 320 windows, three data-split seeds</i>			
align + gate	no	3	0.0028
no-align	yes	3	0.0369
<i>Across data-split seeds, published decoders, same pipeline</i>			
Deep4Net	—	3	0.0265
EEGConformer	—	3	0.0153
EEGNetX	—	3	0.0268
EEGNet	—	3	0.0518
EEGTCNet	—	3	0.0175
FBLightConvNet	—	3	0.0215
ShallowFBCSPNet	—	3	0.0393
TSception	—	3	0.0062

cohort B over three seeds (0.580 against 0.763; lower in 7 of 8 participants, Wilcoxon $p = 0.016$). The cohort A figure comes from one seed and also removes the context module. Against the gate-only arm over three seeds, alignment is worth +0.026 there (Section 6.2), so the size of the inversion

Table 11

Component ablation, cohort A, single seed. Accuracy is the mean over 7 participants with the between-subject SD. Acc@90 is accuracy at 90% coverage; FA/min is spurious activations per minute of standing. All cells use the original estimator so that arms are mutually comparable. **Values await three seeds under the trace-normalised estimator.**

Components	Acc $\pm$ SD	ECE	Acc@90	FA/min
align + gate	0.901 $\pm$ 0.059	0.039	0.876	1.64
align + ctx + gate	0.878 $\pm$ 0.102	0.074	0.837	1.26
align + ctx	0.862 $\pm$ 0.125	0.072	—	1.26
ctx + gate	0.827 $\pm$ 0.096	0.091	0.778	0.73
stem only	0.827 $\pm$ 0.124	0.082	0.823	0.73

Table 12

Artefact control on cohort A. Balanced accuracy of the proposed configuration on the training task, before and after ICA cleaning, with the number of muscle components removed per recording.

Participant	Uncleaned	Cleaned	Change	Muscle removed
sub-01	0.969	0.955	-0.014	2.9
sub-02	0.898	0.914	+0.016	7.1
sub-03	0.955	0.841	-0.114	2.3
sub-04	0.835	0.817	-0.018	1.9
sub-05	0.839	0.893	+0.053	1.3
sub-06	0.673	0.548	-0.124	0.3
sub-07	0.680	0.525	-0.155	2.3
Mean	0.835	0.785	-0.051	

Table 13

Both cohorts, single seed, original estimator. Three-seed cohort B values for *align + gate* are in Table 5.

Components	Cohort A	Cohort B
align + gate	0.901	0.581
align + ctx + gate	0.878	0.626
align + ctx	0.862	0.606
ctx + gate	0.827	0.792
stem only	0.827	0.737

comes mostly from the loss on cohort B. The remainder of the results characterises this inversion.

A protocol-level switch The two cohorts do not require two architectures. The lower bound of Condition 1 acts as a switch whose position is fixed by the recording protocol: with the default estimator memory $\mu = 32/m = 160$ windows, cohort A's $\tau_{\text{blk}} = 18$ satisfies $\tau_{\text{blk}} < \mu$ and cohort B's $\tau_{\text{blk}} = 309$ does not. Enabling alignment exactly when the lower bound is satisfied selects 0.901 on cohort A and 0.792 on cohort B, the best configuration of Table 13 on each at seed 0, from one model and one rule evaluated before any training.

Learnability of the switch The architecture already contains a learnable version of it: the blend weight $g = \sigma(\text{gate})$ in (3) is a trained parameter, so the model is free to close

the gate and bypass alignment. On cohort B it does not, and scores 0.626 at seed 0. The reason is structural and does not lie in optimisation. The gate is trained on the fitting sessions, where the estimator sees interleaved windows and alignment is harmless; the class-tracking failure appears only at test time, when the estimator meets contiguous blocks longer than its memory. **No training signal contains the pathology, so no gradient can close the gate against it.** This is the argument for a protocol-level check instead of an end-to-end one: the quantity that decides whether the layer helps is a property of the evaluation stream, which the fitting distribution does not expose.

6.6. Interventions tested and rejected

Six architectural interventions were tested before the configuration reported here and rejected by their own controls (Table 8). We report them because each control is what distinguishes a mechanism from a correlation, and because two of them looked convincing until the control was run.

Three are multi-seed and test a component the architecture could plausibly have kept.

Temporal smoothing and the attention block Applying an HMM forward filter over the decoder's output improves cohort A by +0.024 over four seeds, and Viterbi decoding by +0.038 (Table 8). Viterbi is reported for reference only: it needs the complete sequence and so cannot run in the causal setting this decoder targets. The control applies the same filter after shuffling window order, which preserves the operation and destroys the temporal structure: it falls to 0.665, **0.113 below** the unsmoothed baseline. A smoothing operation that helped for any reason other than genuine persistence would not behave that way. Participant-paired Wilcoxon signed-rank tests over the seven participants, with seeds averaged within participant and Holm correction across the three comparisons, place the forward filter at +0.024 (better in 6 of 7, $p_{\text{Holm}} = 0.063$), Viterbi at +0.038 (6 of 7, $p_{\text{Holm}} = 0.063$) and the shuffled control at -0.113 (worse in 7 of 7, $p_{\text{Holm}} = 0.047$). **Only the control clears 0.05:** with $n = 7$ the smallest attainable two-sided p is 0.016, so the smoothing gains are consistent in sign but underpowered, while the control's collapse is not. The cross-epoch pathway of Section 6.8 is therefore a real effect even though it costs accuracy.

By contrast, merging the power stem with a within-epoch attention block (Table 8) is improved by 0.024 when the attention is *removed*, across three seeds. The block was dropped.

Replication on an independent architecture We built a separate SPD-manifold decoder in which alignment can be fixed (Euclidean Alignment), learned, or omitted (Table 8). Over three seeds the three variants span 0.814 to 0.823, and omitting alignment entirely costs 0.002, far inside the ± 0.02 to ± 0.03 seed spread. A different backbone, a different alignment parameterisation, and the same conclusion: the alignment stage is not what carries the performance.

Normalisation across published architectures Table 7 repeats the normalisation comparison across seven published decoders. Per-window normalisation is the best choice for none of them and the worst for four, which is the architecture-level counterpart of the representation-level dose–response in Table 6. Table 6 reports the control that retired the amplitude side-channel: supplying a per-window amplitude feature and supplying the same feature with its window assignment shuffled differ by less than 0.01 for five of six models.

Tables 6 and 8 report the representation and architecture sweeps that preceded the design. The tangent-space representation is the strongest single feature family, which is what motivated a covariance-based alignment stage; and no structural change in the 24-configuration sweep reaches 0.06 over the baseline, with widening the network worth less than changing the readout.

The movement-related cortical potential (MRCP) pathway deserves a sentence because it is a physiologically motivated design that failed cleanly. Decoding from the 0.1–3 Hz onset-locked band alone gives 0.604; from the mu and beta ERD band alone, 0.775; combining the two, 0.742. The paired difference between the combination and the better single band is -0.034 , so on single-trial data the low-frequency pathway adds noise, and the architecture uses the ERD band only.

6.7. A rejected explanation for the inversion

One candidate explanation is the power-weighted estimator described in Section 4.1. The defect is real and measurable: on cohort B, where the majority class carries $1.52\times$ the power, the un-normalised running estimate sat at Riemannian distance 0.306 from the walk covariance, which is to say it had become the majority class. The per-window trace normalisation in (3) moves it to 3.899 and 3.194 from the two class covariances.

Correcting it changed decoding by $+0.005$ on cohort B and -0.016 on cohort A. The estimator improved and the task did not follow, so this is therefore not the explanation for the inversion, though the correction is retained, which is independently right. It is the first of five cases in this work where a distribution-level quantity improved without the task following (Section 7).

6.8. The adaptation rate condition

The running estimate tracks whatever is currently streaming. An exponential moving average with momentum m has an effective memory of approximately $1/m$ batches, or NB/m windows at batch size N and B batches. When a protocol presents a class in contiguous blocks longer than that memory, the estimate converges to the currently streaming class, and whitening by it removes the between-class difference. Cohort A's blocks ($\tau_{\text{blk}} = 18$) sit far inside even the fastest memory; cohort B's ($\tau_{\text{blk}} = 309$) do not.

If this mechanism is responsible, slowing adaptation until the memory exceeds a class block should restore cohort B. The memory is approximately 160, 640 and 3200

windows at $m = 0.2, 0.05$ and 0.01 , so the crossing of $\tau_{\text{blk}} = 309$ falls between the first and second settings. Three seeds under the current estimator trace the recovery more finely for *align + gate* (Table 5), against the no-align arm, which does not use the estimator and is averaged over its six runs (0.763). Alignment costs 0.183 at a memory of 160 windows (lower in 7 of 8 participants; Wilcoxon $p = 0.016$), 0.125 at 320 (7 of 8; $p = 0.023$) and 0.068 at 1600 (6 of 8; $p = 0.15$). The cost falls by similar amounts on either side of the block length, so the recovery is gradual and shows no step at $\tau_{\text{blk}} = 309$, and alignment remains harmful at every memory measured. **[The three-seed point at 3200 windows is in progress.]**

At seed 0, every arm that uses alignment gains $+0.090$ to $+0.098$ between memories of 160 and 640 windows (Table 4). Neither arm that omits alignment responds: the no-align arm moves -0.010 , within its 0.0107 run-to-run spread (Table 10), and the stem-only arm reproduces 0.7375 exactly at every setting despite a 160-fold change in the adaptation rate. Because that arm is deterministic, this null is exact. The manipulated variable therefore acts only through the component it controls, which an unrelated confound such as optimisation dynamics or regularisation would not produce.

6.9. Slow adaptation has a cost

On cohort A, whose blocks sit far inside even the fastest memory so that the estimate is already class-mixed, the same change was expected to be neutral. This expectation was registered before the experiment was run. It is not borne out (Table 4).

The optima are opposite, and over three seeds error in either direction costs accuracy: slowing costs 0.050 on cohort A, and on cohort B the default rate gives up 0.115 against a memory of 1600 windows (Table 5). Slower adaptation therefore costs accuracy on cohort A. One reading is a trade-off in which a fast estimate tracks both the streaming class, which is harmful, and session drift, which is beneficial. Measured without a decoder, slowing the estimate also stops it removing drift: on cohort A the reduction in session-to-session distance falls from 45.9 % at $m = 0.2$ to 12.1 % at $m = 0.05$ and 0.2 % at $m = 0.01$. At $m = 0.01$ the estimator's memory, about 3200 windows, is ten times a held-out session of about 313 windows, so this measures how far the estimate can move within the data available and does not isolate a drift timescale (Section 8).

That cost was first measured on *align + ctx + gate*, which is not the configuration this paper proposes. Repeating it on the proposed *align + gate* arm gives a similar figure: slowing to $m = 0.01$ takes cohort A from 0.884 to 0.799 at the first data-split seed under the trace-normalised estimator, a cost of 0.085. Over three seeds the slow arm averages 0.819 ± 0.017 against 0.868 ± 0.026 , a cost of -0.050 with each participant's accuracy averaged over seeds, lower in 7 of 7 participants (Wilcoxon $p = 0.016$, the smallest attainable with seven participants). The cost is -0.085 , -0.056 and -0.008 on the three seeds, so the first-seed figure is the

largest of the three. Both arms use the same estimator version; comparing the slow arm against the original-estimator 0.901 would inflate the cost by mixing estimator versions.

Condition 1 (Admissible rate band). Assume (i) the running estimate is an exponential moving average with effective memory μ windows; (ii) the within-class covariance is stationary over a block; (iii) the protocol presents classes in contiguous blocks of length τ_{blk} ; and (iv) session drift acts as a congruence transform slow on the scale τ_{drift} . Under (i)–(iv) the estimate both avoids tracking the streaming class and tracks session drift only if

$$\tau_{\text{blk}} < \mu < \tau_{\text{drift}}. \quad (4)$$

Where $\tau_{\text{blk}} \geq \tau_{\text{drift}}$ the band is empty and no setting of μ satisfies both requirements. It is a design condition. Its lower bound is tested here empirically, once in advance; its upper bound is not measured, and we give no proof.

Cohort A satisfies the lower bound with a wide margin ($\tau_{\text{blk}} = 18$ against $\mu = 160$), and cohort B's 309-window blocks violate it at the default rate. On cohort B the cost of alignment falls gradually as μ lengthens past τ_{blk} (Section 6.8), so the bound marks where class tracking begins to weaken, and it does not mark a sharp threshold. Whether cohort B has any admissible rate cannot be decided from these data, because its drift timescale could not be measured (Section 8).

Remark 1 (The residual). At the slowest rate measured over three seeds, a memory of 1600 windows, alignment still costs 0.068 on cohort B (0.695 against 0.763). The lower bound does not explain this residual. On cohort A, measured without a decoder, the estimate removes only 2.3 % of session drift at that rate, which would leave estimation noise without a drift correction to offset it, but the same measurement has not been made on cohort B, so the residual is reported as unexplained.

Remark 2 (Offline alignment is exempt). A whole-recording estimate, as in Euclidean Alignment [13], is the maximally slow setting. It therefore never tracks the class, and it never tracks within-recording drift either, which is exactly the capability an in-network version is introduced to add.

The lower bound in (4) is a property of the recording protocol. τ_{blk} is read off the trial structure, so the bound can be checked before a model is fitted, and Section 6.10 tests it on a cohort whose results were not available when the prediction was made. The upper bound was not measured: two label-free estimates of τ_{drift} were attempted, and neither resolved it (Section 8).

6.10. A prediction registered before a third cohort

Condition 1 was formulated after cohort B failed, so on the two cohorts above it is an explanation written with their results in view. To test it in advance, predictions for a

Table 14

Predictions registered before any model was trained on cohort C. Balanced accuracy averaged over all 20 participants, seed 0. Differences are paired within participant.

Prediction	Arms compared	Difference	Threshold	Outcome
P1 alignment costs little	0.783 / 0.785	+0.002	> -0.05	holds
P2 slowing does not help	0.785 / 0.779	-0.006	< +0.05	holds

third cohort were committed to version control before any model was trained on it. The cohort, participants, data split, preprocessing, compared arms and decision thresholds were all fixed at that point.

Cohort C alternates rest and movement about every 4 s. From the published protocol alone a class block holds about five windows, and the loader later measured a median of 5.0. Against an estimator memory of 160 windows at $m = 0.2$, the lower bound of (4) is satisfied by a factor of about thirty, so the estimate cannot converge to either class. Two predictions follow. P1: enabling alignment costs less than 0.05, where the same comparison cost 0.211 on cohort B. P2: slowing adaptation to $m = 0.01$ gains less than 0.05, where it recovered 0.090 on cohort B. The arms compared in P1 differ only in alignment, and those in P2 only in momentum.

Both predictions hold (Table 14). Enabling alignment changes balanced accuracy by +0.002 (higher in 11 participants, lower in 9; Wilcoxon $p = 0.68$), and slowing it changes accuracy by -0.006 (higher in 9, lower in 10; $p = 0.43$). The intervention that recovered 0.090 on cohort B brings no gain here, as the protocol predicted, and enabling alignment shows no harm.

Two further seeds were run after the registered test and are reported separately from it. They agree with both verdicts and sharpen one of them. Enabling alignment changes accuracy by +0.014 (higher in 13 participants, lower in 7; $p = 0.17$), so it again costs nothing, and its small positive trend is not significant. Slowing adaptation costs 0.016 (lower in 15 of 20; $p = 0.003$) and is negative on all three seeds, so on this cohort it carries a small cost. Read after the fact, the sign of slowing is consistent with the lower bound on every cohort. It helps where the estimate tracks the class (cohort B, +0.115 from a memory of 160 to 1600 windows over three seeds) and costs accuracy on both cohorts with short blocks (cohort A, -0.050; cohort C, -0.016). That pattern was observed on the replication seeds and formed no part of the registered prediction.

The test covers the lower bound of Condition 1 and its contrast with cohort B. It does not test the upper bound.

7. Discussion

7.1. Validating a mechanism separately from performance

Domain adaptation and invariance methods are routinely justified by a distribution-level quantity, a divergence, an alignment distance or a nuisance-decodability score, with task performance assumed to follow [64–67]. This work provides five counterexamples within a single system, three of

which we produced by acting on that assumption ourselves (Fig. 6).

1. The adaptation rate was originally selected because it maximised drift reduction (on cohort A, 45.9 % at $m = 0.2$ against 12.1 % at $m = 0.05$ and 0.2 % at $m = 0.01$). That choice costs 0.115 accuracy on cohort B against a memory of 1600 windows, over three seeds (Table 5).
2. The cohort in which more drift was removed (62.5 % against 45.9 %, Table 9) is the cohort in which the layer costs 0.183 accuracy over three seeds (Table 5).
3. A verified correction to a genuine covariance-estimation bias moved decoding by +0.005 and -0.016 (Section 6.7).
4. A prediction registered from our own diagnosis was falsified (Table 4), because the diagnosis accounted only for the harm the estimate can do.
5. The nuisance-leakage probe rises with accuracy whether or not the decoder can access the nuisance, measuring $R^2 = +0.863$ for one that cannot.

Item 4 generalises the rest. Each of these quantities is one-sided by construction: it scores the failure it was designed to detect and carries no term for the capability the intervention removes. Drift reduction measures how much non-stationarity a layer removes and has no term for what the layer destroys. A diagnosis inherited from such a metric inherits its blind spot, which is why our own threshold rule was correct about one direction and silent about the other until we tested it where it predicted no effect. The practical implication is that reporting a reduced divergence, increased invariance or lowered nuisance decodability is evidence about the representation alone, saying nothing about whether the method helps, even when the reduction is large and reproducible.

7.2. Relation to temporally correlated test-time adaptation

The vision literature has identified that test-time adaptation degrades when the test stream is temporally correlated [20–22], and proposes reservoirs and stabilised objectives as remedies. Our result is compatible with that line and adds two things specific to the blocked-protocol case. First, the relevant quantity is a ratio of timescales, measured on the stream, and which side of the threshold a protocol falls on can be predicted in advance, as the third cohort shows (Section 6.10). Second, slowing adaptation is not free. On both cohorts with short blocks it costs accuracy, so remedies that slow adaptation can exchange class tracking for a different loss, which on these cohorts could not be attributed to drift tracking directly.

8. Limitations

Several limitations constrain the interpretation of these results.

No published decoder on cohort B. The published decoders of Table 3 were run on cohorts A and C. [Their

cohort B runs are in progress.] The 0.763 reported on cohort B can therefore be compared with our own arms and with the majority-class floor, and its standing relative to published methods on that cohort is untested.

Seeds. The component ablation and the rate tables are single-seed. The proposed configuration was rerun over three data-split seeds under the trace-normalised estimator, giving 0.868 ± 0.026 (seeds 0.884, 0.882 and 0.839). That seed standard deviation is larger than the 0.010 noise floor of repeated runs at a fixed split (Table 10), and it is concentrated in two participants, whose accuracy ranges over 0.217 and 0.120 across seeds while the other five vary by at most 0.065. The rate effects are therefore reported over three seeds with paired tests (Table 5, Section 6.9). The paired alignment effect on cohort A over the same three seeds is +0.026 ($p = 0.22$) and is negative on one seed, so the benefit of alignment where the lower bound holds is small and unconfirmed at this sample size. The loss where it fails is 0.183 on cohort B over three seeds (lower in 7 of 8 participants; $p = 0.016$). The published decoders were run over the same three seeds on cohorts A and C.

No paired significance tests on accuracy. Per-participant accuracies were not retained for the component ablation and the rate tables, so those comparisons carry dispersion without paired tests. Paired tests are reported wherever per-participant rows exist: the drift results (Table 9), the temporal and normalisation controls, the artefact control (Section 6.3), the three-seed alignment comparison on cohort A (Section 6.2), the comparison with published decoders on cohort C (Section 6.4) and the cohort C predictions (Section 6.10).

Cohort size. Seven, eight and twenty participants. This is typical for exoskeleton BCI studies [1] but small, and the between-subject SD of 0.06 to 0.13 (Table 11) is large relative to the differences between arms.

Best-epoch evaluation. Model selection uses the best epoch by held-out balanced accuracy, which is conventional but upward-biased on small folds. The bias applies equally to all arms and baselines within a pipeline, so comparisons within a pipeline are unaffected, but absolute values should be read as optimistic.

Baseline training. The published decoders were trained with the proposed decoder's settings and no per-model tuning. EEGNet, Deep4Net and EEG-TCNet stayed near chance for several cohort C participants under those settings, so their cohort C results record a training failure and are not evidence about those architectures.

The upper bound is not measured. Two label-free estimates of τ_{drift} were attempted. Block-wise covariance distances within a session are dominated by class alternation where blocks are short, and on cohort A they return about 18 windows, the class-block length. Distances between session mean covariances are already at their plateau at a gap of one session for every cohort A participant, and fall with the gap for two, so a fitted timescale lies below the sampling interval and is not a measurement; cohort B's three trials give two gaps, too few for a fit. The data therefore bound the

drift timescale only from above, at about one session, which does not constrain the adaptation rate. A protocol with more closely spaced recordings is needed to test the upper bound.

Condition 1 is hypothesis-generating. It was formulated after observing cohort B fail. One prediction from it has since been tested out-of-sample, and that test falsified the one-sided form (Table 4), which is how the two-sided form arose. The corrected form has since been tested prospectively on cohort C (Section 6.10), where both registered predictions held. That test covers the lower bound and the contrast with cohort B; the upper bound has not been tested in advance.

Artefact. Retraining after ICA cleaning lowers accuracy by 0.051, concentrated in three participants (Section 6.3). The losses do not follow the amount of muscle activity removed. With seven participants, and with re-referencing part of the cleaning, the control still falls short of establishing that the accuracy is neural. ICLabel is also not trained on gait-related artefact specifically.

Window length differs between cohorts. Cohort B uses 2 s windows against cohort A's 4 s, so the context pathway spans a shorter duration there. This is a confound for the context ablation specifically, though not for the rate condition, which is measured within each cohort.

9. Conclusion

An estimate that adapts at test time tracks whatever is currently streaming. Under a protocol that presents each class in long contiguous blocks, the estimate converges to the current class, and the layer then removes the signal it was inserted to protect. The estimator's memory must therefore outlast a class block. That bound is read from the recording protocol, which makes it a design-time check.

The condition has predicted a new cohort correctly in advance. With class blocks of five windows, enabling alignment cost nothing and slowing it brought no gain, where slowing recovered 0.115 over three seeds on the cohort with long blocks. Where the lower bound holds, the gain from alignment is small and not significant on either cohort that satisfies it: +0.026 over three seeds on cohort A and +0.014 over two replication seeds on cohort C. Where it fails, alignment costs 0.183 over three seeds. The condition is therefore most useful as a design-time guard against that loss.

The immediate open question is the upper bound. Neither cohort here resolves the drift timescale: cohort A's session covariances are already at their plateau one session apart, and cohort B has too few trials to fit one. A protocol with recordings spaced closely enough to sample the rise of drift, and with τ_{blk} between roughly 100 and 200 windows, would place the two bounds close together, and the prediction is that accuracy would peak at an intermediate momentum and fall on both sides. That experiment can be pre-registered from the protocol alone, as the cohort C test was.

A second direction follows from the unexplained residual on cohort B. If a single running estimate cannot simultaneously avoid the class and track the drift, then the two requirements should perhaps be served by two estimators at different rates, with the fast one constrained to the subspace in which drift lives and the slow one supplying the class-neutral reference. Whether such a decomposition is identifiable without labels remains open.

These observations may extend to other online adaptation settings. In five cases here a distribution-level quantity improved without a corresponding task improvement, and in three of those the assumption that it would have already been acted upon. Such measures are one-sided by construction, which suggests that methods justified by a reduced divergence should also report the accuracy cost of that reduction.

CRedit authorship contribution statement

Esm E Moula Chowdhury Abha: Software, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Md Wahiduzzaman Suva:** Software, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Muhammad Hasibur Rashid Chayon:** Supervision, Project administration, Writing – review and editing.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All three datasets are publicly available. All code, configurations and result files, including those of experiments that failed, are in the accompanying repository.

References

- [1] Y. He, D. Eguren, J. M. Azorín, R. G. Grossman, T. P. Luu, J. L. Contreras-Vidal, Brain-machine interfaces for controlling lower-limb powered robotic systems, *Journal of Neural Engineering* 15 (2018) 021004. URL: <https://doi.org/10.1088/1741-2552/aaa8c0>. doi:10.1088/1741-2552/aaa8c0.
- [2] E. López-Larraz, F. Trincado-Alonso, V. Rajasekaran, S. Peres-Nombela, A. J. del Ama, J. Aranda, J. Minguez, A. Gil-Agudo, L. Montesano, Control of an ambulatory exoskeleton with a brain-machine interface for spinal cord injury gait rehabilitation, *Frontiers in Neuroscience* 10 (2016) 359. URL: <https://doi.org/10.3389/fnins.2016.00359>. doi:10.3389/fnins.2016.00359.
- [3] A. H. Do, P. T. Wang, C. E. King, S. N. Chun, Z. Nenadic, Brain-computer interface controlled robotic gait orthosis, *Journal of NeuroEngineering and Rehabilitation* 10 (2013) 111. URL: <https://doi.org/10.1186/1743-0003-10-111>. doi:10.1186/1743-0003-10-111.
- [4] J. L. Contreras-Vidal, R. G. Grossman, NeuroRex: a clinical neural interface roadmap for EEG-based brain machine interfaces to a lower body robotic exoskeleton, *Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)* (2013) 1579–1582. URL: <https://doi.org/10.1109/embc.2013.6609816>. doi:10.1109/embc.2013.6609816.

- [5] A. Presacco, R. Goodman, L. Forrester, J. L. Contreras-Vidal, Neural decoding of treadmill walking from noninvasive electroencephalographic signals, *Journal of Neurophysiology* 106 (2011) 1875–1887. URL: <https://doi.org/10.1152/jn.00104.2011>. doi:10.1152/jn.00104.2011.
- [6] P. Shenoy, M. Krauledat, B. Blankertz, R. P. N. Rao, K.-R. Müller, Towards adaptive classification for BCI, *Journal of Neural Engineering* 3 (2006) R13–R23. URL: <https://doi.org/10.1088/1741-2560/3/1/r02>. doi:10.1088/1741-2560/3/1/r02.
- [7] M. Krauledat, M. Tangermann, B. Blankertz, K.-R. Müller, Towards zero training for brain–computer interfacing, *PLoS ONE* 3 (2008) e2967. URL: <https://doi.org/10.1371/journal.pone.0002967>. doi:10.1371/journal.pone.0002967.
- [8] B. Blankertz, R. Tomioka, S. Lemm, M. Kawanabe, K.-R. Müller, Optimizing spatial filters for robust EEG single-trial analysis, *IEEE Signal Processing Magazine* 25 (2008) 41–56. URL: <https://doi.org/10.1109/msp.2008.4408441>. doi:10.1109/msp.2008.4408441.
- [9] C. Vidaurre, C. Sannelli, K.-R. Müller, B. Blankertz, Machine-learning-based coadaptive calibration for brain–computer interfaces, *Neural Computation* 23 (2011) 791–816. URL: https://doi.org/10.1162/neco_a_00089. doi:10.1162/neco_a_00089.
- [10] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017. URL: <https://arxiv.org/abs/1706.04599>.
- [11] C. K. Chow, On optimum recognition error and reject tradeoff, *IEEE Transactions on Information Theory* 16 (1970) 41–46. URL: <https://doi.org/10.1109/tit.1970.1054406>. doi:10.1109/tit.1970.1054406.
- [12] R. El-Yaniv, Y. Wiener, On the foundations of noise-free selective classification, *Journal of Machine Learning Research* 11 (2010) 1605–1641.
- [13] H. He, D. Wu, Transfer learning for brain–computer interfaces: a Euclidean space data alignment approach, *IEEE Transactions on Biomedical Engineering* 67 (2020) 399–410. URL: <https://doi.org/10.1109/tbme.2019.2913914>. doi:10.1109/tbme.2019.2913914.
- [14] P. L. C. Rodrigues, C. Jutten, M. Congedo, Riemannian Procrustes analysis: transfer learning for brain–computer interfaces, *IEEE Transactions on Biomedical Engineering* 66 (2019) 2390–2401. URL: <https://doi.org/10.1109/tbme.2018.2889705>. doi:10.1109/tbme.2018.2889705.
- [15] O. Yair, M. Ben-Chen, R. Talmon, Parallel transport on the cone manifold of SPD matrices for domain adaptation, *IEEE Transactions on Signal Processing* 67 (2019) 1797–1811. URL: <https://doi.org/10.1109/tsp.2019.2894801>. doi:10.1109/tsp.2019.2894801.
- [16] P. Zanini, M. Congedo, C. Jutten, S. Said, Y. Berthoumieu, Transfer learning: a Riemannian geometry framework with applications to brain–computer interfaces, *IEEE Transactions on Biomedical Engineering* 65 (2018) 1107–1116. URL: <https://doi.org/10.1109/tbme.2017.2742541>. doi:10.1109/tbme.2017.2742541.
- [17] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, T. Darrell, Tent: fully test-time adaptation by entropy minimization, in: *International Conference on Learning Representations (ICLR)*, 2021. URL: <https://arxiv.org/abs/2006.10726>.
- [18] S. Schneider, E. Rusak, L. Eck, O. Bringmann, W. Brendel, M. Bethge, Improving robustness against common corruptions by covariate shift adaptation, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. URL: <https://arxiv.org/abs/2006.16971>.
- [19] Z. Nado, S. Padhy, D. Sculley, A. D’Amour, B. Lakshminarayanan, J. Snoek, Evaluating prediction-time batch normalization for robustness under covariate shift, *arXiv preprint arXiv:2006.10963* (2020). URL: <https://arxiv.org/abs/2006.10963>.
- [20] T. Gong, J. Jeong, T. Kim, Y. Kim, J. Shin, S.-J. Lee, NOTE: robust continual test-time adaptation against temporal correlation, in: *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. URL: <https://doi.org/10.52202/068431-1976>. doi:10.52202/068431-1976.
- [21] S. Niu, J. Wu, Y. Zhang, Z. Wen, Y. Chen, P. Zhao, M. Tan, Towards stable test-time adaptation in dynamic wild world, in: *International Conference on Learning Representations (ICLR)*, 2023. URL: <https://arxiv.org/abs/2302.12400>.
- [22] Q. Wang, O. Fink, L. Van Gool, D. Dai, Continual test-time domain adaptation, in: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. URL: <https://doi.org/10.1109/cvpr52688.2022.00706>. doi:10.1109/cvpr52688.2022.00706.
- [23] R. T. Schirrmeister, J. T. Springenberg, L. D. J. Fiederer, M. Glasstetter, K. Eggersperger, M. Tangermann, F. Hutter, W. Burgard, T. Ball, Deep learning with convolutional neural networks for EEG decoding and visualization, *Human Brain Mapping* 38 (2017) 5391–5420. URL: <https://doi.org/10.1002/hbm.23730>. doi:10.1002/hbm.23730.
- [24] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, B. J. Lance, EEGNet: a compact convolutional neural network for EEG-based brain–computer interfaces, *Journal of Neural Engineering* 15 (2018) 056013. URL: <https://doi.org/10.1088/1741-2552/aace8c>. doi:10.1088/1741-2552/aace8c.
- [25] X. Chen, X. Teng, H. Chen, Y. Pan, P. Geyer, Toward reliable signals decoding for electroencephalogram: a benchmark study to EEGNetX, *Biomedical Signal Processing and Control* 87 (2024) 105475. URL: <https://doi.org/10.1016/j.bspc.2023.105475>. doi:10.1016/j.bspc.2023.105475.
- [26] Y. Roy, H. Banville, I. Albuquerque, A. Gramfort, T. H. Falk, J. Faubert, Deep learning-based electroencephalography analysis: a systematic review, *Journal of Neural Engineering* 16 (2019) 051001. URL: <https://doi.org/10.1088/1741-2552/ab260c>. doi:10.1088/1741-2552/ab260c.
- [27] A. Craik, Y. He, J. L. Contreras-Vidal, Deep learning for electroencephalogram (EEG) classification tasks: a review, *Journal of Neural Engineering* 16 (2019) 031001. URL: <https://doi.org/10.1088/1741-2552/ab0ab5>. doi:10.1088/1741-2552/ab0ab5.
- [28] Y. Song, Q. Zheng, B. Liu, X. Gao, EEG Conformer: convolutional transformer for EEG decoding and visualization, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 31 (2023) 710–719. URL: <https://doi.org/10.1109/tnsre.2022.3230250>. doi:10.1109/tnsre.2022.3230250.
- [29] T. M. Ingolfsson, M. Hersche, X. Wang, N. Kobayashi, L. Cavigelli, L. Benini, EEG-TCNet: an accurate temporal convolutional network for embedded motor-imagery brain–machine interfaces, in: *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2020. URL: <https://doi.org/10.1109/smc42975.2020.9283028>. doi:10.1109/smc42975.2020.9283028.
- [30] A. Barachant, S. Bonnet, M. Congedo, C. Jutten, Multiclass brain–computer interface classification by Riemannian geometry, *IEEE Transactions on Biomedical Engineering* 59 (2012) 920–928. URL: <https://doi.org/10.1109/tbme.2011.2172210>. doi:10.1109/tbme.2011.2172210.
- [31] A. Barachant, S. Bonnet, M. Congedo, C. Jutten, Classification of covariance matrices using a Riemannian-based kernel for BCI applications, *Neurocomputing* 112 (2013) 172–178. URL: <https://doi.org/10.1016/j.neucom.2012.12.039>. doi:10.1016/j.neucom.2012.12.039.
- [32] M. Congedo, A. Barachant, R. Bhatia, Riemannian geometry for EEG-based brain–computer interfaces: a primer and a review, *Brain–Computer Interfaces* 4 (2017) 155–174. URL: <https://doi.org/10.1080/2326263x.2017.1297192>. doi:10.1080/2326263x.2017.1297192.
- [33] F. Lotte, L. Bougrain, A. Cichocki, M. Clerc, M. Congedo, A. Rakotomamonjy, F. Yger, A review of classification algorithms for EEG-based brain–computer interfaces: a 10 year update, *Journal of Neural Engineering* 15 (2018) 031005. URL: <https://doi.org/10.1088/1741-2552/aab2f2>. doi:10.1088/1741-2552/aab2f2.
- [34] D. Wu, Y. Xu, B.-L. Lu, Transfer learning for EEG-based brain–computer interfaces: a review of progress made since 2016, *IEEE Transactions on Cognitive and Developmental Systems* 14 (2022) 4–19. URL: <https://doi.org/10.1109/tcds.2020.3007453>. doi:10.1109/tcds.2020.3007453.

- [35] Y. Sun, X. Wang, Z. Liu, J. Miller, A. A. Efros, M. Hardt, Test-time training with self-supervision for generalization under distribution shifts, in: Proceedings of the 37th International Conference on Machine Learning (ICML), 2020. URL: <https://www.semanticscholar.org/paper/a436e41141a9d9c5dfc44d16fade67b9a3b76778>.
- [36] M. Boudiaf, R. Mueller, I. Ben Ayed, L. Bertinetto, Parameter-free on-line test-time adaptation, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022. URL: <https://doi.org/10.1109/cvpr52688.2022.00816>.
- [37] M. P. Naeini, G. F. Cooper, M. Hauskrecht, Obtaining well calibrated probabilities using Bayesian binning, in: AAAI Conference on Artificial Intelligence, 2015. URL: <https://doi.org/10.1609/aaai.v29i1.9602>. doi:10.1609/aaai.v29i1.9602.
- [38] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. V. Dillon, B. Lakshminarayanan, J. Snoek, Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift, in: Advances in Neural Information Processing Systems (NeurIPS), 2019. URL: <https://arxiv.org/abs/1906.02530>.
- [39] Y. Geifman, R. El-Yaniv, SelectiveNet: a deep neural network with an integrated reject option, in: Proceedings of the 36th International Conference on Machine Learning (ICML), 2019, pp. 2151–2159. URL: <https://arxiv.org/abs/1901.09192>.
- [40] Y. Geifman, R. El-Yaniv, Selective classification for deep neural networks, in: Advances in Neural Information Processing Systems (NIPS), 2017. URL: <https://arxiv.org/abs/1705.08500>.
- [41] J. T. Gwin, K. Gramann, S. Makeig, D. P. Ferris, Electroocortical activity is coupled to gait cycle phase during treadmill walking, *NeuroImage* 54 (2011) 1289–1296. URL: <https://doi.org/10.1016/j.neuroimage.2010.08.066>. doi:10.1016/j.neuroimage.2010.08.066.
- [42] M. Seeber, R. Scherer, J. Wagner, T. Solis-Escalante, G. R. Müller-Putz, EEG beta suppression and low gamma modulation are different elements of human upright walking, *Frontiers in Human Neuroscience* 8 (2014) 485. URL: <https://doi.org/10.3389/fnhum.2014.00485>. doi:10.3389/fnhum.2014.00485.
- [43] T. P. Luu, S. Nakagome, Y. He, J. L. Contreras-Vidal, Real-time EEG-based brain–computer interface to a virtual avatar enhances cortical involvement in human treadmill walking, *Scientific Reports* 7 (2017) 8895. URL: <https://doi.org/10.1038/s41598-017-09187-0>. doi:10.1038/s41598-017-09187-0.
- [44] J. Wagner, T. Solis-Escalante, P. Grieshofer, C. Neuper, G. Müller-Putz, R. Scherer, Level of participation in robotic-assisted treadmill walking modulates midline sensorimotor EEG rhythms in able-bodied subjects, *NeuroImage* 63 (2012) 1203–1211. URL: <https://doi.org/10.1016/j.neuroimage.2012.08.019>. doi:10.1016/j.neuroimage.2012.08.019.
- [45] G. Pfurtscheller, F. H. Lopes da Silva, Event-related EEG/MEG synchronization and desynchronization: basic principles, *Clinical Neurophysiology* 110 (1999) 1842–1857. URL: [https://doi.org/10.1016/s1388-2457\(99\)00141-8](https://doi.org/10.1016/s1388-2457(99)00141-8). doi:10.1016/s1388-2457(99)00141-8.
- [46] G. Pfurtscheller, C. Brunner, A. Schlögl, F. H. Lopes da Silva, Mu rhythm (de)synchronization and EEG single-trial classification of different motor imagery tasks, *NeuroImage* 31 (2006) 153–159. URL: <https://doi.org/10.1016/j.neuroimage.2005.12.003>. doi:10.1016/j.neuroimage.2005.12.003.
- [47] M. Sugiyama, M. Krauledat, K.-R. Müller, Covariate shift adaptation by importance weighted cross validation, *Journal of Machine Learning Research* 8 (2007) 985–1005. URL: <https://doi.org/10.5555/1314498.1390324>. doi:10.5555/1314498.1390324.
- [48] S. Ioffe, C. Szegedy, Batch normalization: accelerating deep network training by reducing internal covariate shift, in: Proceedings of the 32nd International Conference on Machine Learning (ICML), 2015, pp. 448–456. URL: <https://arxiv.org/abs/1502.03167>.
- [49] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, in: Advances in Neural Information Processing Systems (NIPS), 2017. URL: <https://arxiv.org/abs/1706.03762>.
- [50] H. Phan, F. Andreotti, N. Cooray, O. Y. Chén, M. De Vos, SeqSleepNet: end-to-end hierarchical recurrent neural network for sequence-to-sequence automatic sleep staging, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 27 (2019) 400–410. URL: <https://doi.org/10.1109/tnsre.2019.2896659>. doi:10.1109/tnsre.2019.2896659.
- [51] H. Phan, O. Y. Chén, M. C. Tran, P. Koch, A. Mertins, M. De Vos, XSleepNet: multi-view sequential model for automatic sleep staging, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44 (2021) 5903–5915. URL: <https://doi.org/10.1109/tpami.2021.3070057>. doi:10.1109/tpami.2021.3070057.
- [52] A. Supratak, H. Dong, C. Wu, Y. Guo, DeepSleepNet: a model for automatic sleep stage scoring based on raw single-channel EEG, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 25 (2017) 1998–2008. URL: <https://doi.org/10.1109/tnsre.2017.2721116>. doi:10.1109/tnsre.2017.2721116.
- [53] G. Schalk, D. McFarland, T. Hinterberger, N. Birbaumer, J. Wolpaw, BCI2000: A general-purpose brain-computer interface (BCI) system, *IEEE Transactions on Biomedical Engineering* 51 (2004) 1034–1043. URL: <https://doi.org/10.1109/tbme.2004.827072>. doi:10.1109/tbme.2004.827072.
- [54] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, H. E. Stanley, PhysioBank, PhysioToolkit, and physionet, *Circulation* 101 (2000). URL: <https://doi.org/10.1161/01.cir.101.23.e215>. doi:10.1161/01.cir.101.23.e215.
- [55] A. Gramfort, M. Luessi, E. Larson, D. A. Engemann, D. Strohmeier, C. Brodbeck, R. Goj, M. Jas, T. Brooks, L. Parkkonen, M. Hämäläinen, MEG and EEG data analysis with MNE-Python, *Frontiers in Neuroscience* 7 (2013) 267. URL: <https://doi.org/10.3389/fnins.2013.00267>. doi:10.3389/fnins.2013.00267.
- [56] D. P. Kingma, J. Ba, Adam: a method for stochastic optimization, in: International Conference on Learning Representations (ICLR), 2015. URL: <https://arxiv.org/abs/1412.6980>.
- [57] R. T. Schirrmester, L. Gemein, K. Eggensperger, F. Hutter, T. Ball, Deep learning with convolutional neural networks for decoding and visualization of EEG pathology, *IEEE Signal Processing in Medicine and Biology Symposium (SPMB)* (2017). URL: <https://doi.org/10.1109/spmb.2017.8257015>. doi:10.1109/spmb.2017.8257015, braindecode library.
- [58] Y. Ding, N. Robinson, S. Zhang, Q. Zeng, C. Guan, TSception: capturing temporal dynamics and spatial asymmetry from EEG for emotion recognition, *IEEE Transactions on Affective Computing* 14 (2023) 2238–2250. URL: <https://doi.org/10.1109/taffc.2022.3169001>. doi:10.1109/taffc.2022.3169001.
- [59] A. Salami, J. Andreu-Perez, H. Gillmeister, EEG-ITNet: an explainable inception temporal convolutional network for motor imagery classification, *IEEE Access* 10 (2022) 36672–36685. URL: <https://doi.org/10.1109/access.2022.3161489>. doi:10.1109/access.2022.3161489.
- [60] P. Ablin, J.-F. Cardoso, A. Gramfort, Faster independent component analysis by preconditioning with Hessian approximations, *IEEE Transactions on Signal Processing* 66 (2018) 4040–4049. URL: <https://doi.org/10.1109/tsp.2018.2844203>. doi:10.1109/tsp.2018.2844203.
- [61] L. Pion-Tonachini, K. Kreutz-Delgado, S. Makeig, ICLabel: An automated electroencephalographic independent component classifier, dataset, and website, *NeuroImage* 198 (2019) 181–197. URL: <https://doi.org/10.1016/j.neuroimage.2019.05.026>. doi:10.1016/j.neuroimage.2019.05.026.
- [62] T. Castermans, M. Duvinage, G. Cheron, T. Dutoit, About the cortical origin of the low-delta and high-gamma rhythms observed in EEG signals during treadmill walking, *Neuroscience Letters* 561 (2014) 166–170. URL: <https://doi.org/10.1016/j.neulet.2013.12.059>. doi:10.1016/j.neulet.2013.12.059.
- [63] J. E. Kline, H. J. Huang, K. L. Snyder, D. P. Ferris, Isolating gait-related movement artifacts in electroencephalography during human walking, *Journal of Neural Engineering* 12 (2015) 046022. URL: <https://doi.org/10.1088/1741-2560/12/4/046022>. doi:10.1088/1741-2560/12/4/046022.

- [64] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, V. Lempitsky, Domain-adversarial training of neural networks, *Journal of Machine Learning Research* 17 (2016) 1–35. URL: https://doi.org/10.1007/978-3-319-58347-1_10. doi:10.1007/978-3-319-58347-1_10.
- [65] M. Long, Y. Cao, J. Wang, M. I. Jordan, Learning transferable features with deep adaptation networks, in: *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, 2015. URL: <https://arxiv.org/abs/1502.02791>.
- [66] B. Sun, K. Saenko, Deep CORAL: correlation alignment for deep domain adaptation, in: *ECCV Workshops*, 2016. URL: https://doi.org/10.1007/978-3-319-49409-8_35. doi:10.1007/978-3-319-49409-8_35.
- [67] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, J. W. Vaughan, A theory of learning from different domains, *Machine Learning* 79 (2010) 151–175. URL: <https://doi.org/10.1007/s10994-009-5152-4>. doi:10.1007/s10994-009-5152-4.

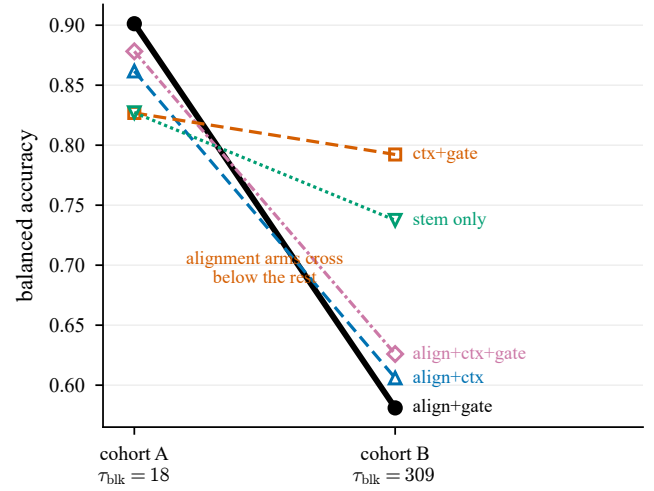


Figure 4: The same five arms on both cohorts, single seed. Solid heavy black is the proposed configuration; the other alignment-using arms are dashed in colour, and the two alignment-free arms are dashed and dotted. The three alignment arms cross below both alignment-free arms, so the ordering inverts instead of compressing.

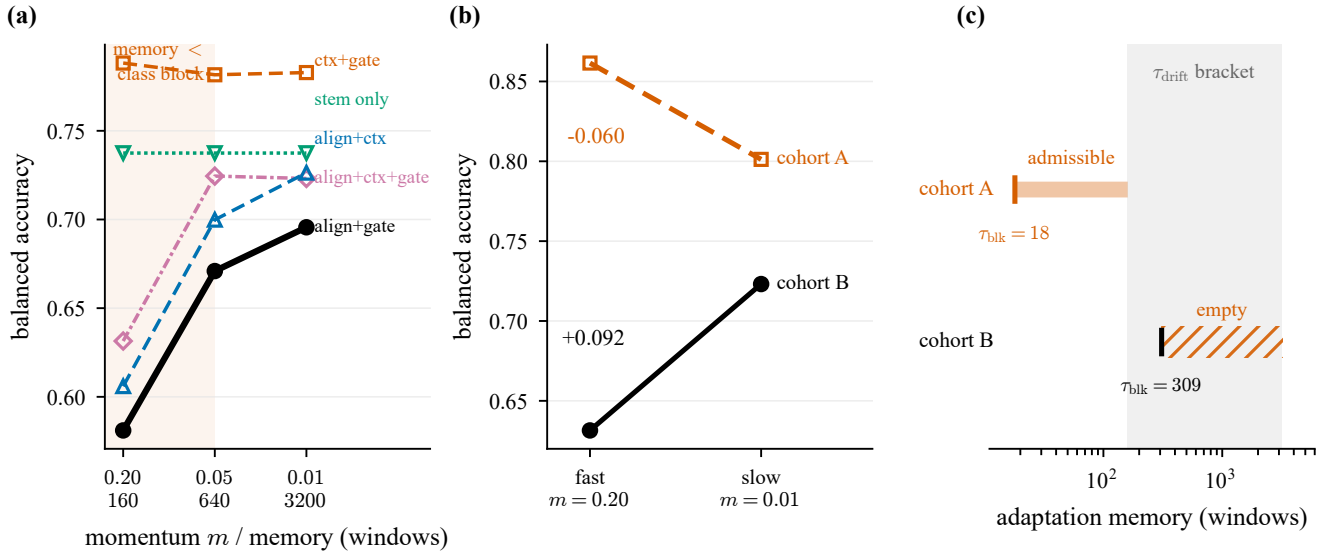


Figure 5: The adaptation rate condition. (a) Cohort B, five arms, three momenta, single seed; the shaded region is where the estimator memory is shorter than a class block. Measurement points are marked because the sweep is sparse. (b) The same change applied to both cohorts at matched estimator version, giving opposite signs. (c) The admissible band of Condition 1. Its lower bound is τ_{blk} ; its upper bound, τ_{drift} , was not measured on these cohorts (Section 8). [Panel (c) to be redrawn without a drawn upper bound; panel (a) to be regenerated with three seeds.]

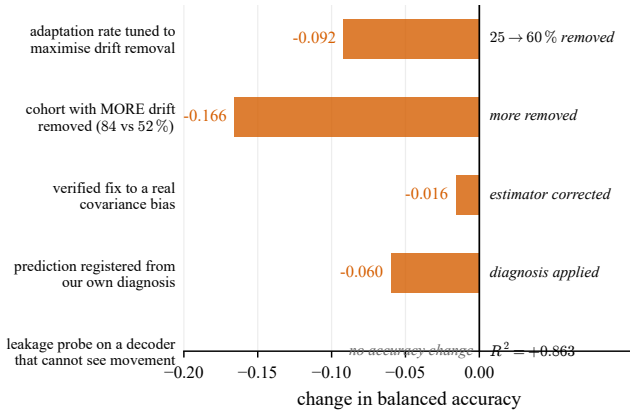


Figure 6: Five cases in which a distribution-level quantity improved (right, in blue) and the task did not follow (left, in red). The final row has no accuracy change and therefore no bar: the leakage probe rises with accuracy whether or not the decoder can access the nuisance.